

What are Escape Sequences?

Escape sequences are special characters used inside a C++ string to perform a specific action or display special characters.

An escape sequence usually starts with a **backslash** (\).

Common Escape Sequences

Escape Sequence	Name	Purpose
\n	Newline	Moves the cursor to the next line
\t	Tab	Adds a horizontal tab space
\\	Backslash	Displays a backslash \
\"	Double Quote	Displays a double quotation mark "
\'	Single Quote	Displays a single quotation mark '
\r	Carriage Return	Moves the cursor to the beginning of the current line

1. Newline – \n

What is \n?

The \n escape sequence is used to **move the cursor to the next line**.

It is similar to using endl.

Question

Write a C++ program that prints two lines of text using the newline escape sequence \n.

Example

```
#include <iostream>
using namespace std;

int main() {

    cout << "Hello World!\n";
    cout << "Welcome to C++.";

    return 0;
}
```

Output

```
Hello World!  
Welcome to C++.
```

Understand the Code

```
cout << "Hello World!\n";
```

The `\n` moves the cursor to the next line. Therefore, the second `cout` statement starts on a new line.

Remember

```
\n → New line
```

2. Tab – `\t`

What is `\t`?

The `\t` escape sequence inserts a **horizontal tab space**.

It is useful for creating simple columns in output.

Question

Write a C++ program that displays a list of items and their prices using `\t` to separate the columns.

Example

```
#include <iostream>  
using namespace std;  
  
int main() {  
  
    cout << "Item\tPrice\n";  
    cout << "Apple\t100\n";  
    cout << "Mango\t150\n";  
  
    return 0;  
}
```

Output

```
Item    Price  
Apple   100
```

Mango 150

Understand the Code

```
cout << "Apple\t100\n";
```

- `\t` adds a tab space between `Apple` and `100`.
- `\n` moves the cursor to the next line.

Remember

`\t` → Tab space

3. Backslash – `\\`

What is `\\`?

The backslash character `\` has a special meaning in C++.

If we want to **display an actual backslash**, we use two backslashes:

```
\\
```

Question

Write a C++ program that displays a Windows file path containing backslashes.

Example

```
#include <iostream>
using namespace std;

int main() {
    cout << "File path: C:\\Users\\Student\\Documents";

    return 0;
}
```

Output

```
File path: C:\Users\Student\Documents
```

Understand the Code

In the program:

```
C:\\Users\\Student\\Documents
```

Each `\\` represents **one actual backslash** in the output.

Remember

```
\\ → Displays \
```

4. Double Quote – `\"`

What is `\"`?

A double quotation mark `"` normally marks the beginning or end of a string.

If we want to display a double quotation mark **inside a string**, we use:

```
\"
```

Question

Write a C++ program that prints a sentence containing double quotation marks.

Example

```
#include <iostream>
using namespace std;

int main() {
    cout << "He said, \"C++ is fun!\"";
    return 0;
}
```

Output

```
He said, "C++ is fun!"
```

Understand the Code

```
\ "C++ is fun!\ "
```

The `\ "` tells C++ to display a quotation mark instead of treating it as the end of the string.

Remember

```
\ " → Displays "
```

5. Single Quote – `\ '`

What is `\ '`?

The `\ '` escape sequence can be used to display a **single quotation mark**.

Question

Write a C++ program that prints a character enclosed in single quotation marks.

Example

```
#include <iostream>
using namespace
```